



Transportation Analytics Project for CityTransit Corp

Introduction

CityTransit Corp seeks proposals for a six-week capstone project in transportation data engineering. As urban congestion grows, transportation analytics becomes critical: the market is projected to rise from \$22.02 billion in 2024 to \$120.66 billion by 2034. Teams will build data pipelines to process traffic and transit datasets, aiming to optimize routes and improve mobility. This project mimics an urban data team analyzing mobility patterns to enhance city transportation planning.

Project Scope and Objectives

Teams will simulate solving urban mobility problems using public datasets:

- **Data Ingestion:** Gather open transportation data. Potential sources include city traffic sensor feeds (e.g., DOT open data portals), public transit ridership logs, and ride-sharing trip records (e.g., NYC Taxi & Limousine Commission data). Integrate multiple data types (e.g., bus GPS, bike-share, subway entries).
- **Data Cleaning:** Handle GPS noise and missing logs, normalize coordinate systems (e.g., ensure consistent map projections), and synchronize timestamps (e.g., convert all to UTC). Merge heterogeneous sources on common keys (time, location).
- **Data Transformation & Storage:** Create a spatio-temporal data warehouse (partitioned by date and region) or data lake capable of handling large trip datasets. Compute derived fields (e.g., trip durations, average speeds).
- **Analysis & Modeling:** Perform route optimization or demand forecasting: for example, use historical data to model peak traffic flows and suggest optimized transit schedules. Apply clustering to identify congestion hotspots, or simulate optimal paths (e.g., using shortest-path algorithms).
- **Visualization:** Build dashboards (e.g., congestion heatmaps, interactive maps of traffic volume) to convey insights like average travel times or incident impacts. Provide recommendations for improving transit operations.



By working with transportation data, students will address real-world problems like traffic management and multi-modal integration. Optimizing flows can significantly enhance urban mobility.

Phases and Weekly Deliverables

The project is organized into weekly phases:

- **Phase 1 (Weeks 1–2): Data Collection and Preparation**
 - *Week 1:* Identify and acquire transportation datasets (e.g., traffic counts, transit schedules, trip records). Document data schemas (fields like latitude/longitude, timestamps) and coordinate systems. **Deliverable:** Data acquisition log and raw data summaries.
 - *Week 2:* Clean and unify data: remove erroneous GPS points, fill data gaps (e.g., due to sensor outages), and ensure consistent time zones. **Deliverable:** Cleaned and integrated dataset.
- **Phase 2 (Weeks 3–4): Pipeline Development & Feature Engineering**
 - *Week 3:* Build ETL pipelines to import cleaned data into a spatial database (e.g., PostGIS) or big data store. Automate data loading (e.g., nightly batch jobs). **Deliverable:** Working data ingestion pipeline and data storage schema.
 - *Week 4:* Engineer features: compute metrics like average speed on roads, passenger load per hour, or travel time distributions. Optimize storage (create spatial and temporal indexes). **Deliverable:** Feature-engineered tables and indexing strategy documentation.
- **Phase 3 (Weeks 5–6): Modeling, Visualization, & Reporting**
 - *Week 5:* Analyze and model: apply clustering (e.g., identify congestion clusters) or forecasting (e.g., predict ridership for next week). Use routing algorithms (e.g., Dijkstra's) for path optimization. **Deliverable:** Models or analyses with performance evaluation.
 - *Week 6:* Visualize results: create map-based dashboards (e.g., using GIS tools or Python libraries) showing congestion heatmaps or optimized routes. Prepare final report and presentation. **Deliverable:** Interactive



visualization and final presentation slides.

Mentors will advise on correct spatial analysis methods. Teams should collaborate closely, with responsibilities like data engineer (pipelines), GIS specialist, and analyst.

Use of Generative AI

Use AI tools to streamline the workflow:

- **ChatGPT (OpenAI):** Generate code for spatial data processing (e.g., cleaning GPS data) and explain geospatial concepts (e.g., how clustering works in mapping).
- **GitHub Copilot:** Help write code in Python or SQL for data transformations and spatial queries, such as iterating over grid cells or formatting GIS queries.
- **Gemini (Google):** Query Gemini for examples of transit data analyses or to assist with using Google's mapping APIs in code.
- **Claude (Anthropic):** Use Claude to articulate analysis findings in clear prose (e.g., summarizing the effect of peak hours on congestion).
- **LangChain:** Employ LangChain to chain geo-based prompts, such as refining a routing strategy by iteratively prompting an LLM with map data, or automating multi-step spatial analyses.

Document AI usage, noting if Copilot streamlined writing spatial queries or if ChatGPT helped troubleshoot coordinate conversion.

Expected Outcomes

Teams will deliver:

- A robust pipeline ingesting and storing multi-modal transportation data.
- Clean, integrated mobility datasets (GPS, ridership, schedules).
- Analytical models of traffic or transit demand and actionable insights (e.g., recommended route adjustments).



- Visual tools (maps, charts) illustrating mobility patterns and improvements.
- A discussion of AI tools used (e.g., ChatGPT for code, Gemini for ideas) and how they aided development.

These outcomes prepare students for smart city and transportation analytics roles, highlighting data-driven planning benefits.

Deliverables

- Git repository with all code and documentation.
- Cleaned and feature-engineered transport datasets with descriptions.
- Phase reports including intermediate analyses (e.g., maps of cleaned vs. raw data).
- Analytical model results (e.g., predicted congestion levels, suggested routes) with evaluation metrics.
- Mapping visualizations/dashboards (GIS tools or Python) showing network traffic and optimizations.
- Final presentation detailing methodology and recommendations.

Proposal Submission Requirements

Submit a PDF containing:

- **Project Goals:** Goals (e.g., reduce average commute times).
- **Data Plan:** Types of data (traffic counts, GPS), sources, and access methods.
- **Technical Approach:** GIS tools, databases (e.g., PostGIS), programming languages (e.g., Python, R).
- **Team and Schedule:** Roles and week-by-week plan (1–6).



- **Relevant Experience:** (e.g., GIS, transportation systems).

Ensure privacy (anonymize any sensitive location data) and adherence to data usage policies.

Evaluation Criteria

Projects will be assessed on:

1. **Data Handling:** Completeness and efficiency of data pipeline for large spatial datasets.
2. **Analytical Insight:** Quality of traffic models and optimization (e.g., reduction in travel time).
3. **AI Integration:** Use of AI tools (e.g., ChatGPT to explain algorithms).
4. **Innovation:** Novel methods (e.g., multi-modal integration, routing solutions).
5. **Communication:** Clarity of visualizations and reports for city planners.